

Design of a Modular Anesthesia System for Greater Efficiency in Care

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2.750 Medical Device Design

Abstract

Existing infusion and bolus dose workflows create operational bottlenecks that limit the adoption of optimal anesthesia strategies for patients. In particular, the practice of “multi-modal” anesthesia reduces side effects and improves patient recovery, but requires the use of a wider variety of medications. Current bolus workflows require providers to manually prepare each medication, requiring additional labor for anesthesia providers, considerable medication and single-use plastics waste, and more opportunities for administration error.

In this paper, we present a prototyped multi-use medication dispenser designed for ergonomics, medicine administration error prevention, and waste reduction to incentivize optimal anesthesia strategies for patients. Features of the dispenser include: a per-day medication setup and a per-patient tubing set, a modular design for reconfiguration by the user, and a streamlined UI with built-in guardrails.

We compare key aspects of the design with existing infusion systems to provide quantitative metrics for: dosing accuracy, waste generation, and time savings. We also qualitatively justify the user interface design using interaction studies from existing workflows.

1. Introduction

1.1 Challenges in Anesthesia Practices

Surgeons rely on anesthesiologists to maintain patient unconsciousness, prevent pain, suppress movement, and manage physiological instability throughout an operation. These effects are typically achieved through the administration of multiple medications with distinct pharmacologic roles, including sedatives, analgesics, neuromuscular blockers, and vasoactive drugs. During surgery, anesthesiologists must continuously monitor the patient’s condition and dynamically adjust medication delivery to maintain the desired level of anesthesia and stability. Current medication administration systems introduce considerable setup complexity and workflow demands in the operating room.

Anesthesiologists typically give medications through one or more intravenous lines using a combination of manual bolus injections and programmable infusion pumps. While many anesthetics can rely on just one or two pumps, anesthesia for complex surgery often requires eight to ten separate infusions. While infusion pumps allow for hands-free medication delivery, they must be individually primed and programmed prior to the procedure, something anesthesia providers spend an average of 6 minutes on per procedure. Other medications are given in discrete bolus doses, which are injected via syringe into the IV line. While bolus medications don’t require pump setups, they must be manually drawn from vials into syringes, often prior to the start of the procedure, so they can be ready at a moment’s notice.

Although there have been significant advances in the practice of anesthesiology, innovation in medication administration has not kept up with the field. Over the past decade, there has been a new focus on “multi-modal” anesthesia, which involves giving upwards of 15 medications concurrently rather than the “usual recipe” of around 10 medications. Using more medications allows for a smaller dose of any single drug and, therefore, fewer side effects from any single drug ([Egan & Svensen, 2018](#)). While this is an appealing approach, it is often impractical because it requires considerable setup time and may create significant waste since the actual amount of a particular medication used is proportionally smaller. Technological advances that streamline anesthesia provider workflows are crucial in enabling these improved clinical practices.

In parallel with increasing clinical complexity, modern surgical care is also shaped by strong economic and operational forces. There is considerable pressure to improve operational efficiency to reduce healthcare costs. Ambulatory surgical centers (ASCs) host over 80% of surgeries in the US, rather than traditional hospital operating rooms, a number that is only projected to grow (Daniell, 2026). These facilities prioritize high procedural throughput and rapid operating room turnover in order to maintain financial viability and maximize patient access to care (Daniell, 2026; DataPoints Issue 26: Trends in Ambulatory Surgical Centers in Massachusetts, 2024). As a result, efficiency in perioperative workflows (including anesthesia preparation and medication administration) has become an important operational consideration. Processes that require extensive manual preparation, such as drawing medications into individual syringes, assembling infusion lines, and configuring multiple pumps, can increase setup time and contribute to delays between cases. In high-volume surgical environments, even small inefficiencies in anesthesia preparation and medication delivery may accumulate to produce substantial impacts on cost, staffing demands, and operating room utilization.

Despite advances in anesthetic pharmacology, the infrastructure used to prepare and administer anesthetic medications remains largely manual. This limitation becomes increasingly problematic as anesthetic practice shifts toward more complex, multi-drug regimens while surgical environments simultaneously demand greater efficiency and faster operating room turnover. Together, these trends highlight an opportunity for innovation in the architecture of anesthesia drug delivery systems. Improved solutions that simplify medication preparation, maintain sterility, and allow flexible real-time control of drug administration could meaningfully reduce workflow burden while supporting safer and more efficient anesthetic care.

1.2 Transferable Infusion Setups

Transferable infusion setups allow for patient-to-patient transfer of some or all components of the anesthesia delivery mechanism. Current anesthesia practices do not implement this technique due to concerns regarding sterility and contamination. Research in the use of serial non-return valves, NRVs, for contamination prevention could enable improvements.

As far back as 1996, the foundational work of Gretzinger et. al. supports using serial NRVs for contamination prevention. They performed testing of various NRVs with appropriate back pressure and identified valve cracking pressure as a requirement for efficacy. Greater cracking pressure led to sterile outcomes, with the key threshold of >3psi. NRVs above this threshold cracking pressure are available in standard medical component catalogs, yet transferable infusion setups have been slow to adopt this paradigm.

Some more recent works have argued against the use of serial NRVs and cast doubt on their ability to prevent contamination, which has slowed adoption (Eichler et al., 2004; Ellger et al., 2011; Nandy et al., 2017). We performed an analysis of these negative results and identified consistent error sources that offer an explanation. Studies by Nandy and Ellger explicitly used insufficient valve cracking pressure. Ellger even comments on valve selection, stating that the selected valves were insufficient to prevent leakage, but did not move towards higher cracking pressures (Ellger et al., 2011). A study by Eichler does not state cracking pressure, but they do state their rate. According to the work of Radke et. al., a second source of risk is flow rate, with greater flow reducing contamination risk. Radke used a minimum flow rate of 10ml/hr, which had a higher contamination risk than the higher flow infusions, and Eichler used a maximum flow rate of 2ml/hr, falling well below this guideline. Another possible contamination source is tubing handling (Karlsson et al., 2024; Rebergue & Boisgard, 2015). Karlsson et al. performed comparison testing of single-use versus multi-use infusion sets and found similar contamination risk, which they suggest was caused by handling. Rebergue and Boisgard took this further and performed contamination analysis based on handling technique (Rebergue & Boisgard, 2015). They were able to successfully identify an aseptic technique. Current literature has identified inconsistent results with serial NRVs' efficacy, but we have identified inconsistencies in implementation that offer an explanation.

When cracking pressure, flow rate, and handling technique are considered together, serial NRVs can work to prevent contamination (Gretzinger et al., 1996; Rens et al., 2025). Consider a recent study by Rens et. al., which conducted trials with the appropriate FDA guidance on microbial ingress testing with 51 patients across 12-hour periods and found no contaminants, viral or bacterial, in any sample (Rens et al., 2025). Rens considered valve selection and aseptic technique. Ellgers' work was used to motivate correct valve selection, but they do not state cracking pressure explicitly. They also do not explicitly mention flow rate as a consideration, which is an acknowledged limitation of this comparison. We argue that this stands as a success in

both cases. If it were with a limited flow rate, only 2 of 3 requirements were sufficient for sterility. If it were with a higher flow rate, we return to the base case of meeting all three requirements. We argue for the more restrictive case of satisfying all 3 requirements, which satisfies both conditions.

Serial one-way valves enable transferable infusion setups, allowing for new automation techniques that require less labor than traditional anesthesia administration. Per-day medicine setup and automated medication delivery provide opportunities for interface improvements that weren't possible with disposable mechanisms.

2. Problem Statement

This paper describes a concept user interface and proof-of-concept prototype of a multi-medication dispenser that streamlines anesthesiologists' drug setup and delivery methods. The device leverages recent developments in transferable infusion setups to reduce the need for single-use syringes by pumping medications directly from the vial into the IV line. This device will reduce operating room turnover, decrease medication and plastic waste, and improve anesthetic efficiency and safety for surgical patients. Improving the overall efficiency of discrete bolus dose delivery expands the capabilities of anesthesia providers, enabling practices such as multi-modal anesthesia, which are currently impractical.

3. Design

Intraoperative anesthesia delivery is among the most cognitively demanding tasks in the clinical setting. Anesthesiologists must continuously monitor patient vitals, manage multiple drug infusions simultaneously, and respond to physiological changes in real time, all while operating under significant time pressure. Conventional anesthesia delivery systems fragment this workflow across multiple discrete devices: anesthesia machines where gases are administered, separate infusion pumps for each drug, and physical vials and syringes requiring manual preparation and dose calculation. This fragmentation increases the number of discrete interactions required per clinical decision and introduces multiple opportunities for error.

Figure 1 shows the modular anesthesia system (MAS). MAS was designed to address these limitations through interface consolidation and workflow simplification. The primary design objective for the integration between modular hardware and user interface is to reduce the time and cognitive effort required to initiate, adjust, and monitor drug delivery. Specific goals for the hardware include modularity, which allows anesthesiologists to retain the same amount of flexibility in drawing up medication in syringes while still implementing a process that reduces waste. Specific goals for the interface design include consolidating all drug control and monitoring into a single unified touchscreen interface, minimizing the number of interactions

required to execute common clinical tasks such as rate adjustment and bolus delivery, and ensuring that critical status information is legible at a glance from across the operating room.

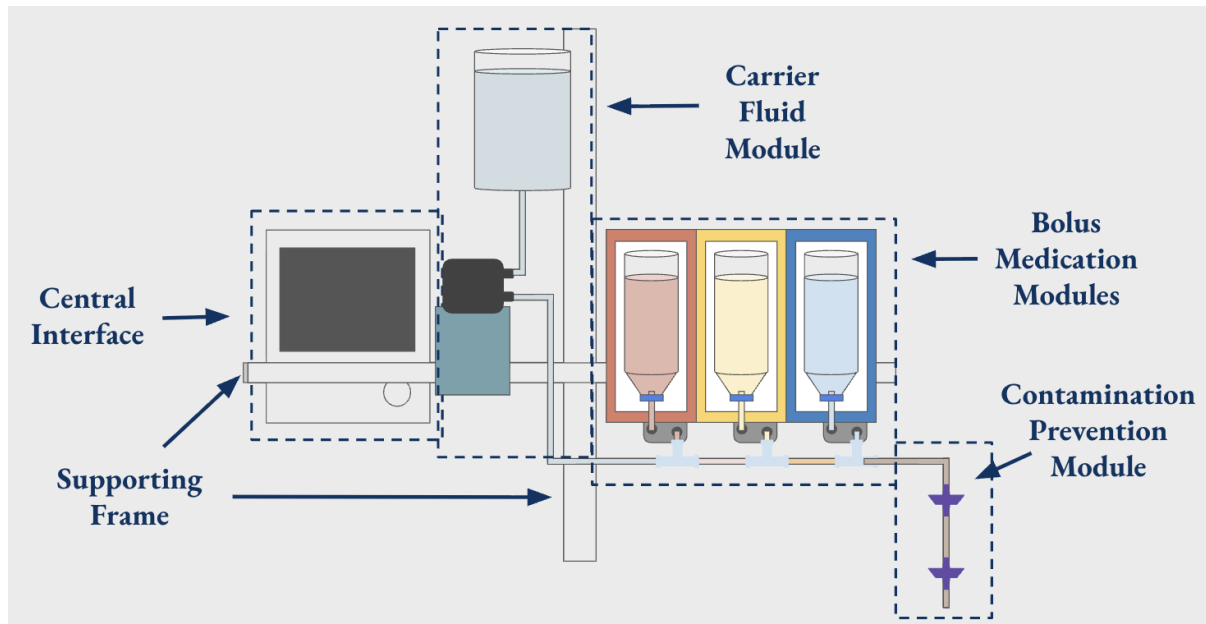


Figure 1: MAS module layout overview

3.1 UI Design

The interface is further constrained by the physical realities of the operating room environment. Controls must be operable by clinicians wearing surgical gloves, which reduces the fine motor precision. Any changes to drug administration must be executed immediately upon the clinician's input with negligible latency. All consequential actions, such as starting an infusion, delivering a bolus, or pausing a drug, must require deliberate confirmation to prevent accidental activation. These constraints shaped fundamental decisions about input modality, visual hierarchy, and interaction flow throughout the design process.

The MAS interface was developed in direct collaboration with Dr. Neil Rens, who serves as the primary clinical domain expert throughout the design process. Interface mockups reflect the task sequence of an experienced clinician managing an intraoperative case, which forms the foundation of the interface architecture. Figure 2 references the drug library within MAS, showing how drug color coding follows the ISO 26825 international standard for anesthetic agent labeling, assigning yellow to induction agents, red to relaxants, blue to opioids, and violet to vasopressors. Reversal agents corresponding to each category are further distinguished through a diagonal hash pattern overlaid on the class color, consistent with established conventional labeling systems (ISO Standard No. 26825:2008).

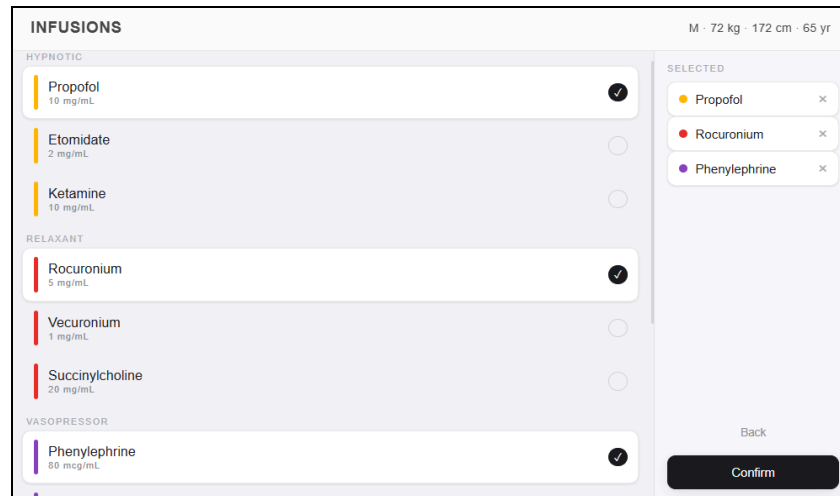


Figure 2: Drug library interface

The MAS interface employs a deliberate split between two input modalities: a touchscreen for navigation and categorical selection, and a physical rotary encoder for numerical value entry and confirmation. This division reflects the distinct ergonomic demands of each task type. Touchscreen interaction is well-suited for discrete selections, where the target is a clearly bounded element and precision is not critical. Numerical entry, by contrast, requires fine continuous adjustment under conditions where tactile feedback is particularly valuable. The detented clicks of a rotary encoder provide confirmation of each increment for precise, intuitive control. All numerical inputs in the system are shown in Figures 3 through 5 (patient data, infusion rates, and bolus doses) are entered exclusively through the encoder, with the touchscreen reserved for navigation and confirmation.

DOSE SETUP M - 72 kg - 172 cm - 65 yr

Propofol 10 mg/mL	Rocuronium 5 mg/mL	Phenylephrine 80 mcg/mL
INFUSION RATE 100 mcg/kg/min	INFUSION RATE 0.4 mcg/kg/min	INFUSION RATE 20 mcg/min
BOLUS PRESETS - MCG 50 100 150 + Add	BOLUS PRESETS - MCG 20 40 60 + Add	BOLUS PRESETS - MCG 40 80 160 + Add
Back	Start Infusion	

INFUSION ACTIVE M - 72 kg - 172 cm - 65 yr RUNNING

Propofol 10 mg/mL	Rocuronium 5 mg/mL	Phenylephrine 80 mcg/mL
RUNNING	PAUSE	RUNNING
INFUSION RATE 100 mcg/kg/min	INFUSION RATE 0.4 mcg/kg/min	INFUSION RATE 20 mcg/min
BOLUS - MCG 50 100 150 200 Custom	BOLUS - MCG 20 40 60 Custom	BOLUS - MCG 40 80 160 Custom
PAUSE	PAUSE	PAUSE
Stop All		

Figure 3-5: Patient data input, drug dose setup and dynamic control interface

Several interface decisions were made specifically to reduce the risk of consequential errors during use. Bolus delivery follows a two-step flow: the clinician selects an amount, reviews it in a confirmation window, confirms delivery, and receives a subsequent acknowledgement screen closing the feedback loop. Infusion rate changes incorporate a two-second debounce delay so that the pump does not respond until the encoder has been stationary, preventing transient fluctuations while the clinician is mid-adjustment. Paused drug columns are visually dimmed with a gray wash across the entire column, providing an unambiguous status legible from across the room without requiring close inspection of the display.

3.2 Hardware Design

Because drawing up medication from vials and discarding the remaining volume after each surgery is wasteful, MAS aims to make medication available for transfer between patients. To do so, MAS is designed to use large-volume vials that last an entire day of surgery. The mechanical design takes into account current hospital processes and equipment. The resulting components of this system are: the supporting frame, the carrier fluid module, bolus medication modules, and the contamination prevention modules.

The supporting frame is a modified IV pole. This ensures that standard hospital IV bags can still be hung from the hooks at the top of the pole. A 3D-printed mount allows two poles to come together with a protruding flange large enough to mount a bar of 80/20 extrusion. This extrusion acts as a rail system for our modules to slide on and off of, and is the base of the modularity offered by this device.

Next, the carrier fluid module is designed simply to hold a large Kamoer KMH400 pump such that IV tube management is simple and effective, including Luer lock connections for ease of changing the tubing at the end of the day. This module is also intended to slide on and off the railing for full modularity, but should be placed upstream of the medication vials to successfully carry any bolus doses administered. The main IV tube line downstream of the pump modules utilizes small-diameter tubing to allow for quicker medicine administration. We used a 1.8ml tube set, which can be fully flushed in less than 0.5 seconds to deliver medication.

The bolus medication modules feature an open resting place for ease of medication access. This module, shown in Figure 6, is designed to house a vented spike and a 100mL vial. It is worth noting that 100mL is just for demonstration and proof of concept prototyping. The system can be scaled to hold volumes of vials that are appropriate for surgeries within a 12-hour period. A slight press fit was designed to keep the spike in place, and compliant grippers keep the vial in place once slotted in. In the case that a medication vial is emptied, the process for replacing it is shown in Figure 7. The vial can be easily slotted upwards, flipped over, and unspiked, and the new vial can be spiked, flipped, and slotted in. Each module is also designed with a motor mount for the Kamoer KMH100 on the opposite side of the rail in order to minimize the time it takes for medication to go through the system and reach the patient. At the start of each day, the tubing from the vial to the main carrier fluid line should be primed.

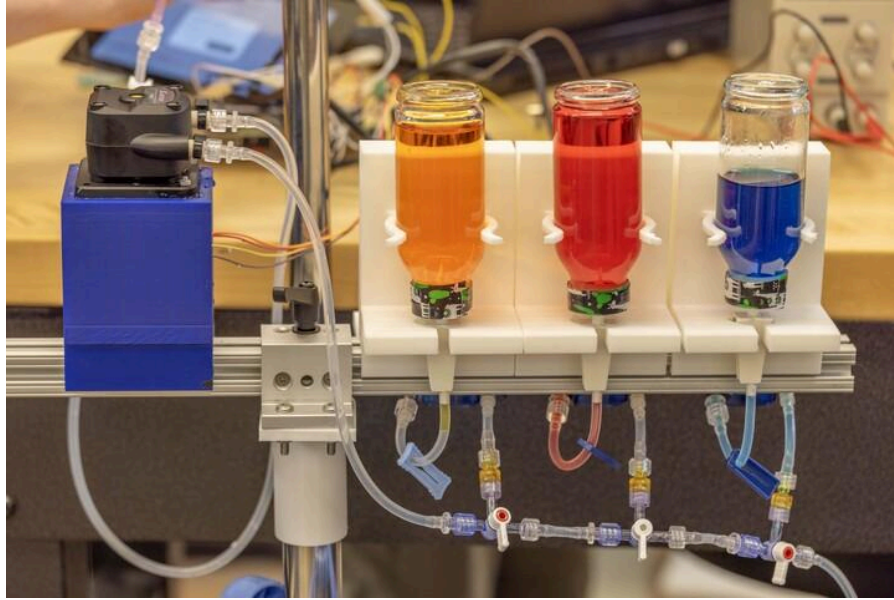


Figure 6: Real world system of pump modules.

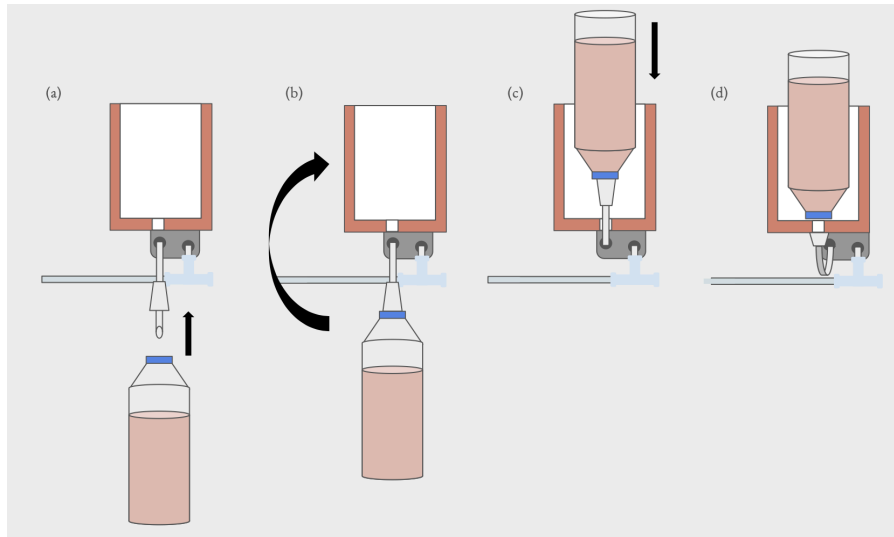


Figure 7: Process diagram of medication vial loading. (a) Spike the vial. (b) Flip it into the module. (c) Slot the vial into place. (d) Loaded vial.

The contamination modules feature two one-way valves in series. These valves have a cracking pressure of 5 psi and a breaking pressure of at least 13 psi based on empirical testing with a Pendotech UX-59900-27 pressure sensor shown in Figure 8. Because the average venous blood pressure of a human is about 0.04-0.29 psi, these valves ensure no backflow from the patient up the line to the medication. They also exceed the >3 psi recommendation for contamination prevention from Gretzinger for contamination prevention, allowing for patient-to-patient transfer.

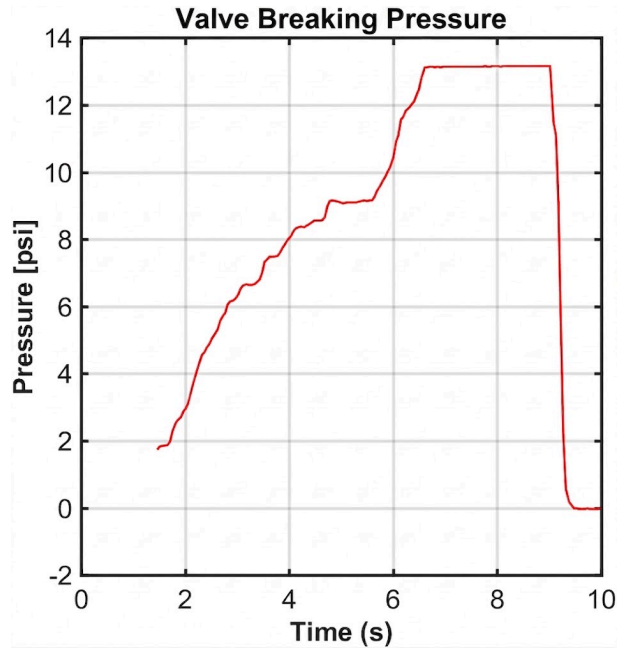


Figure 8: Burst Pressure test simulating high-pressure backflow through MAS

4. Methods

4.1 Accuracy Testing

We performed initial accuracy testing for MAS using a modified version of IEC 60601-2-24 Accuracy Test for Pumps type 3* in (*IEC 60601-2-24*, n.d.). The test setup, shown below in Figure 9, involves pumping water through the system and into a container placed on the scale, and measuring the mass of the output. A major deviation from the IEC 60601-2-24 setup was the use of regular tap water instead of ISO class III water for medical use. The bolus delivery accuracy tests require the measurement of 25 discrete bolus doses at both the minimum and maximum bolus volumes. Twenty-five trials of the minimum bolus dose (0.1cc) were conducted, while only 10 trials of the maximum bolus dose (10cc) were conducted due to time constraints. After each bolus dose, the scale was re-balanced to give a new measurement.

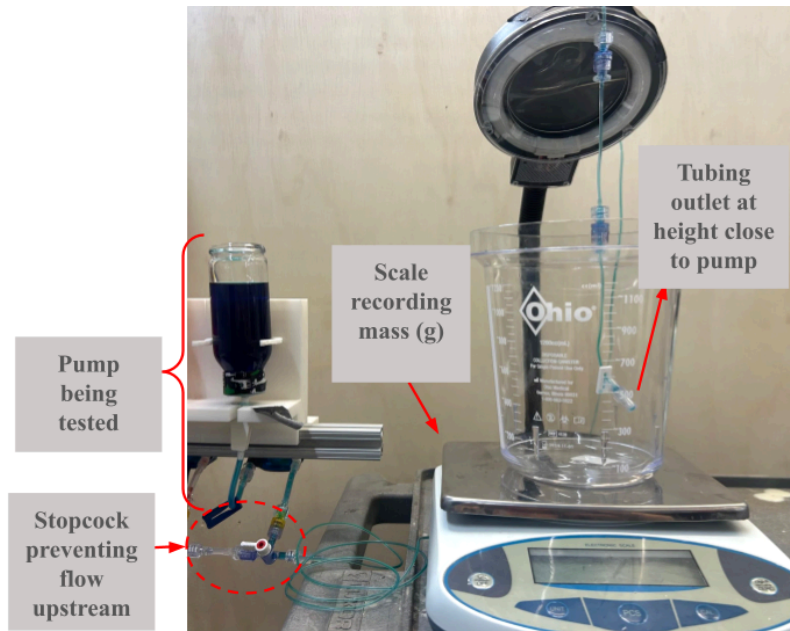


Figure 9: Test setup for volumetric infusion pumps as described in IEC 60601. The test apparatus measures the dosing accuracy by weighing samples of the infusion pump output at specified doses.

4.2 Time Testing

The authors performed testing of vial loading and medicine administration with 25 trials for 3 different users. Time was measured using a digital stopwatch. For vial loading, the time measured was from vial pickup to drug selected from the library in the interface. For medicine administration, the time measured started from when the bolus dose was inputted to when the medicine began dispensing.

5. Results and Discussion

5.1 Dosing Accuracy

Traditional methods of bolus administration require hand-drawn syringes. Around 29% of syringe medications are misdrawn, and the concentration falls $\pm 10\%$ outside of the desired. 18% of preparations deviated by $\pm 20\%$ from the declared dose. Further, in 4% of cases, discrepancies exceeded 100%, which could suggest that dilution or dosing calculations were performed incorrectly (Stucki).

Current methods for medication preparation reveal a high rate of discrepancy. Over 7,000 Americans die yearly due to medication errors (Stucki). These errors are currently ascribed to human error in drawing up the medication and initial volume-to-draw calculations.

MAS has the advantage of computer control, allowing for high accuracy and guard rails to streamline calculations. Accuracy tests were performed to validate its performance, and the results are featured in Table 1.

	Volume [cc]	Mean [cc]	Standard Deviation	Percent Deviation
Min Dose	0.10	0.10	0.01	10.00%
Max Dose	10.00	10.00	0.02	0.20%

Table 1: Bolus Medication accuracy testing

There were a few limitations of our testing. First, our scale was only accurate to 0.01g, which is the standard deviation we acquired for low dosing. Second, fluid self-adhesion at low doses would dominate and turn the fluid deposition into discrete droplets with larger steps than our stated pump step size. This would not be a concern in real-world conditions, as the carrier fluid flow would dominate and deposit the bolus in a continuous flow.

For low-dose bolus administration, our standard deviation was on the order of the accuracy of our scale. Therefore, we likely overestimate true system variability. Say we have a desired dose of 0.1cc delivered by MAS, assuming a normal distribution, a mean of 0.1cc, and a standard deviation of 0.01cc. From this, we expect 31.7% to fall outside the $\pm 10\%$ bounds and 4.55% to fall outside the $\pm 20\%$ bounds. Manual bolus administration has 29% and 18% outside the $\pm 10\%$ and $\pm 20\%$ bounds, respectively. This demonstrates comparable $\pm 10\%$ bounds, but significantly tighter $\pm 20\%$ bounds for MAS compared to manual.

In contrast, say we have a desired dose of 10cc delivered by MAS, assuming a normal distribution, a mean of 10cc, and a standard deviation of 0.02cc. From this, we expect approximately 0% to fall outside the $\pm 10\%$ bound. The dominant error source appears approximately constant in magnitude rather than proportional to delivered volume. We suggest this is the fluid self-adhesion previously mentioned.

Low dose values have comparable measured accuracy, and high dose values are significantly more accurate. Additionally, the MAS system has narrower distributions with a reduced probability of large deviations. The reduced chance of catastrophic failure and comparable accuracy demonstrate the advantage of moving away from manual bolus administration.

5.2 Waste Generation

Anesthesia drug wastage, particularly intravenous agents, is a hidden yet significant financial and environmental cost in the OR.

For manual bolus medication administration, wasted drugs amount to \$10.86/case, and the syringes cost \$1.10/case (Weinger, 2001). Or, a total per case cost of \$11.96. Worth noting, this underreports cost as it does not consider partially used vials discarded within the day or syringes that were discarded at the end of the day. These limitations were accepted to preserve the blind nature of the study and prevent anesthesiologists from altering their behavior (Weinger, 2001).

MAS is designed to minimize this waste by making the system available for all-day use and minimizing single-use elements. It does require two serial NRVs and a tube set for disposable elements. We will not consider the tube set as this cost is maintained between manual administration and MAS. Appropriate NRVs cost as little as 50 cents each with high volume orders (*SCV06252 | Nordson MEDICAL*, n.d.). Additionally, MAS does not require syringes, which provides a cost reduction of \$1.10/case.

While it is difficult to quantify the exact amount of medicine waste with MAS without additional testing, its medicine waste is generated per day, instead of per case, offering an inherent advantage. For the purpose of comparison, we will say that at the end of the day, MAS will still have a case's worth of drugs. In other words, \$10.86 per day. We can justify this estimate by acknowledging that the medication container sizes can be selected appropriately for the planned case load.

Given between 6-10 cases per day, we estimate MAS generates between \$13.86 and \$15.86 worth of waste. Manual bolus administration generates between \$71.76-\$119.60 worth of waste. MAS therefore saves \$57.9 at minimum, and \$103.74 for a reasonably busy day's use.

5.3 Time Savings

Time spent on drawing up bolus medications to prep for operations is a significant factor in anesthesiologist workflow. Anesthesiologists prepare about 10 medicine syringes for an average case (Whitaker & Whitaker, 2024). It takes providers approximately 35 seconds to draw up a completely unprepared emergency drug (Fraind et al., 2002). It then takes an additional 13 seconds to administer the drug, which only accounts for when the medication enters the line (Fraind et al., 2002). We will assume that each medication drawn is unique. MAS benefits from using multiple of the same medication, as it only needs to mount a medication once for any number of doses, so this comparison does not favor MAS

The MAS system takes 7.85 ± 0.24 s to load a new medication and 2.25 ± 0.12 s to administer a loaded drug. This result is not only an improvement in an emergency loading situation, but it is a per-day time cost for medication loading, providing further benefits. Table 2 provides a breakdown of the time improvement MAS provides to the current emergency drug administration practice.

	Current Workflow Time Estimate [s] (Fraind et al.)	Mean [s]	Margin of Error [s] (95% CI)	Difference [s]
Load Medication	35	7.85	0.24	27.15
Bolus	13	2.25	0.12	10.75

Table 2: Time estimation of MAS tasks. The current time taken to administer emergency medication as presented in Fraind et al. is compared to the time estimates of MAS obtained via the Time Testing Method [Section 4.2].

5.4 User Experience

While modular infusion pump systems exist on the market, their intended use is mainly for continuous-rate infusions over longer periods of time, rather than discrete bolus doses. The cognitive demand from operating complex interfaces in fast-paced environments is a frequent cause for error from providers, posing serious risks like medication overdose (Wan et al., 2026). The development of patient-transferrable infusion sets creates the opportunity to reimagine the traditional peristaltic or syringe pump found in the OR. Most infusion pumps have a similar interface consisting of a central screen with soft keys to navigate through different functions. A recent study on smart pump usability for ICU nurses found an overwhelming preference for touchscreen displays as they were found to have clearer displays, be more intuitive to operate, be faster to program, and not be prone to wearing out like a mechanical button (Blake et al., 2022). Another common complaint among infusion pump users is difficulty navigating between different settings and modes (Wan et al., 2026). The MAS improves upon these standards by utilizing a touchscreen display that creates an intuitive interface and minimizes functionalities not necessary for the administration of anesthesia. This reduces the cognitive load on anesthesia providers, allowing them to be more focused on patient care and avoid mistakes.

6. Conclusions

MAS demonstrates that meaningful improvements in anesthesia drug delivery are achievable through the integration of transferable infusion hardware, automated pump control, and a consolidated user interface. Across the dimensions evaluated (dosing accuracy, waste generation, setup time, and user experience), MAS compares favorably to conventional manual administration and offers clear pathways toward measurable clinical and operational benefit.

Dosing accuracy at high bolus columns approaches near perfect precision, and at minimum doses, the system shows tight distributions. The per-day medication setup model converts drug waste from a per-case cost to a per-day cost. The modular hardware design enables

rapid medication loading and reconfiguration, reducing the preparation burden that currently deters the adoption of complex multi-modal anesthetic regimens. The interface design reflects a deliberate alignment with the cognitive and ergonomic constraints of the operating room. Together, these decisions reduce the interaction burden on providers without sacrificing the deliberateness that patient safety demands.

7. Implications and Applications

This work represents a proof-of-concept prototype rather than a clinical system, and several limitations warrant acknowledgement. Accuracy testing was conducted with tap water rather than clinical fluids, the scale resolution constrained low-dose measurements, and a formal usability evaluation with anesthesia providers has not yet been conducted. Future work should pursue validation under clinically representative conditions, integration with electronic health record systems for patient data transfer, and expanded contamination testing across a broader range of flow rates and valve configurations to further strengthen the evidence base for transferable infusion setups.

The confluence of rising surgical volume, pressure toward multi-modal anesthesia, and operational demands of ambulatory surgical centers creates a clear need for innovation in anesthesia drug delivery infrastructure. MAS offers a concrete step toward a system that is simultaneously safer, less wasteful, and more compatible with the complex pharmacologic strategies that modern anesthesia practice increasingly demands.

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